CPSE to FDE Readiness Assessment Scorecard

Purpose

This scorecard is the shared evaluation instrument used at every stage of the CPSE to FDE readiness assessment. Gate reviewers use it to score artifacts before the panel. Panel members use it during the discussion to update scores based on live discussion. The final scorecard is returned to the candidate as feedback regardless of outcome. The competency areas are grounded in an AI focused subset of the Google Well Architected Framework https://docs.cloud.google.com/architecture/framework.

Scoring Model

Each sub-category is scored on a 0–3 scale. The levels are cumulative. A higher score implies the lower levels are also met.

Score	Level	Definition
0	**Not Demonstrated**	Topic not addressed adequately in artifacts or discussion.
1	**Awareness**	Can identify the concern and articulate why it matters. Understands the problem space but has not applied it.
2	**Competent**	Can articulate a concrete approach with trade-offs in the subcategory. May have partially implemented. Can answer follow-up questions about alternatives and limitations.
3	**Proficient**	Has deep production-level or production intended experience in this area. Can discuss edge cases, failure modes, and alternatives from

Score	Level	Definition
		first-hand knowledge of **at least one** of the listed aspects.

Scoring guidance: Score what the candidate *demonstrates*, not what you believe they could do. Artifacts set the floor; panel discussion can raise a score but cannot compensate for absent artifacts entirely (a candidate who discusses security eloquently but submitted nothing demonstrating it should not exceed a 2 in that competency area).

Sub-category descriptions indicate the scope of each category. They are not exhaustive checklists; panelists may explore related topics within a category's scope.

AI/ML Engineering

Sub-category	Description	Score (0–3)	Notes
Agentic & Multi-Agent Systems	Agent orchestration patterns, tool/function calling, planning and reasoning loops, memory and state management, multi-agent coordination, cognitive architectures, and integrating agents with legacy/external APIs.		
Retrieval & Data Engineering for AI	RAG architecture patterns, embedding models and vector databases, chunking and indexing strategies, hybrid search, grounding and citation, data pipelines for AI, parsing complex document formats (PDF/HTML tables), and citation/grounding.		
Model Selection, Tuning & Optimization	Model assessment and selection (quality, latency, cost, context), tuning, quantization, distillation, prompt engineering, structured output enforcement (JSON mode), and inference optimization.		

Sub-category	Description	Score (0–3)	Notes
	Data engineering (preparation, labeling, synthetic generation), architecture selection (encoder/decoder).		
LLM Ops and Evaluation	Evaluation methodology for generative AI, quality metrics (faithfulness, relevance, coherence), eval dataset design, human vs. automated evaluation, regression detection, Observability (tracing/logging).		
Domain-Applied AI/ML Expertise	Translation of vertical-specific business KPIs into AI/ML objectives and architectures. Domain-specific feature engineering, data handling, and regulatory compliance, and privacy regulations (e.g., HIPAA, FINRA). Integration of AI solutions with industry-standard legacy systems and workflows.		

Scoping and Documentation

Sub-category	Description	Score (0–3)	Notes
Problem Definition	Identifying and articulating the business problem, translating ambiguity into a technical opportunity, documenting what success unlocks for the customer		
Technical Scope & Constraints	Defining technical scope, identifying constraints and assumptions, documenting system boundaries		

Sub-category	Description	Score (0–3)	Notes
	and the architectural approach prior to build		
Stakeholder Alignment & Success Criteria	Evidence of stakeholder alignment, defined success criteria and acceptance criteria, phased delivery plan or architectural roadmap		
System Design Artifacts	Architecture diagrams, data flow diagrams, sequence diagrams		
Decision Records	ADRs, documenting trade-offs, alternatives considered, rationale		
API Documentation	OpenAPI specs, integration guides, SDK documentation		
Operational Documentation	Deployment guides, runbooks, onboarding docs, troubleshooting guides		

Security, Privacy & Compliance

Sub-category	Description	Score (0–3)	Notes
Authentication & Authorization	Service accounts, IAM, API key scoping, OAuth/OIDC, Least Privilege		
Infrastructure & Network Security	VPC design, network segmentation, private endpoints, zero-trust architecture, service perimeters, firewall policies		
Data Protection & Privacy	Encryption at rest/transit, PII handling, data classification, DLP, Secrets management		

Sub-category	Description	Score (0–3)	Notes
AI-Specific Security	Prompt injection mitigation, output filtering, guardrails, model access controls, content safety		
Compliance & Governance	Audit logging, data residency, regulatory requirements, policy enforcement		

Reliability & Resilience

Sub-category	Description	Score (0–3)	Notes
Availability Design	Redundancy, failover, health checks, SLO/SLA definition, distributed systems patterns (consistency models, partitioning, multi-region, inter-service communication)		
Observability	Structured logging, metrics, distributed tracing, AI-specific monitoring (latency, token usage, quality)		
Failure & Recovery Testing	Failure injection, red teaming, disaster recovery validation, backup/restore testing, resilience testing under degraded conditions		
Graceful Degradation	Fallback strategies, circuit breakers, retry policies, timeout handling		

Performance & Cost Optimization

Sub-category	Description	Score (0–3)	Notes
Scalability & Elasticity	Horizontal and vertical scaling strategies, autoscaling policies, load balancing, API throughput design, HPC/GPU cluster scaling, and scalable inference.		
Resource Efficiency	Compute/GPU right-sizing, autoscaling strategies, spot/preemptible usage		
AI Cost Management	Model selection trade-offs, response caching, batching, memory and embeddings, data and context management, token optimization		

Operational Excellence

Sub-category	Description	Score (0–3)	Notes
CI/CD & Deployment	Pipeline design, deployment strategies (blue/green, canary), rollback automation		
Infrastructure as Code	Terraform/Pulumi, Cloud Build, reproducible environments, environment parity		
AI Lifecycle Management	Model and agent versioning, A/B testing, prompt and agent configuration management, experiment tracking, agent operational tooling		

Sub-category	Description	Score (0–3)	Notes
Testing & Quality Engineering	Unit, integration, e2e testing, load and performance testing, test automation, test coverage		

Designing for Change

Sub-category	Description	Score (0–3)	Notes
Modularity & Abstraction	Loose coupling, interface contracts, model swappability, dependency injection		
Configuration Management	Environment separation, feature flags, externalized config		
API Design & Versioning	Backward compatibility, contract-first design, schema evolution		
Extensibility	Plugin patterns, event-driven architecture, webhook/callback support		

How This Scorecard Is Used

Stage	Who Scores	What's Scored	Purpose
Gate Review	Gate reviewer(s)	Artifacts only	Determine gate outcome (reject / panel)
Panel Discussion	Panel members	Artifacts + live discussion	Final readiness assessment

Stage	Who Scores	What's Scored	Purpose
Feedback	Process owners	Final merged scorecard	Returned to candidate with gap analysis